

MULTISCALE MODELLING OF CD9 CONFORMATIONAL DYNAMICS AND MEMBRANE REMODELING

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ABSTRACT

Extracellular vesicles are nanoscale membrane-bound carriers involved in intercellular communication and in multiple physiological and pathological processes [1]. The causal, mechanistic links among EV membrane composition, size, membrane protein organization, and biological function remain largely to be elucidated. Tetraspanins are key markers and structural regulators of extracellular vesicles, contributing to the formation of a dynamic interaction platform known as the tetraspanin web [2,3]. Here, we focus on CD9, a member of the tetraspanin family, and develop a coarse-grained Martini 3 [4] multistate model able to represent distinct conformational states of the protein. We use atomistic and coarse-grained simulations in a multiscale workflow to characterize CD9 conformational variability and protein–lipid interactions, in a homogeneous POPC membrane and in an EV-like asymmetric membrane [5]. We quantify the effect of a single and multiple CD9s on membrane mechanics using the sticky-tape set-up [6] and local mechanical descriptors. We find that CD9 molecules perturb bilayer shape, curvature, and elastic properties. In particular, we quantify how CD9 modulates local tilt and splay elastic moduli, while separately probing curvature generation and curvature sensing as distinct membrane-remodelling mechanisms. Our results reveal a decoupling between these two behaviours: CD9 preferentially partitions to positively curved membrane regions, in line with experimental evidence for curvature sensing by tetraspanins [7], while its curvature-generating effect shifts the membrane towards negative curvature. This separation is consistent with a mechanical framework in which CD9 locally softens the bilayer, reducing the elastic cost of membrane deformation and allowing curvature sensing and curvature generation to emerge as partially independent responses [8]. Together, these analyses provide a molecular-level view of how CD9 can couple protein conformational dynamics, lipid composition, and membrane mechanics in EV-like bilayers.

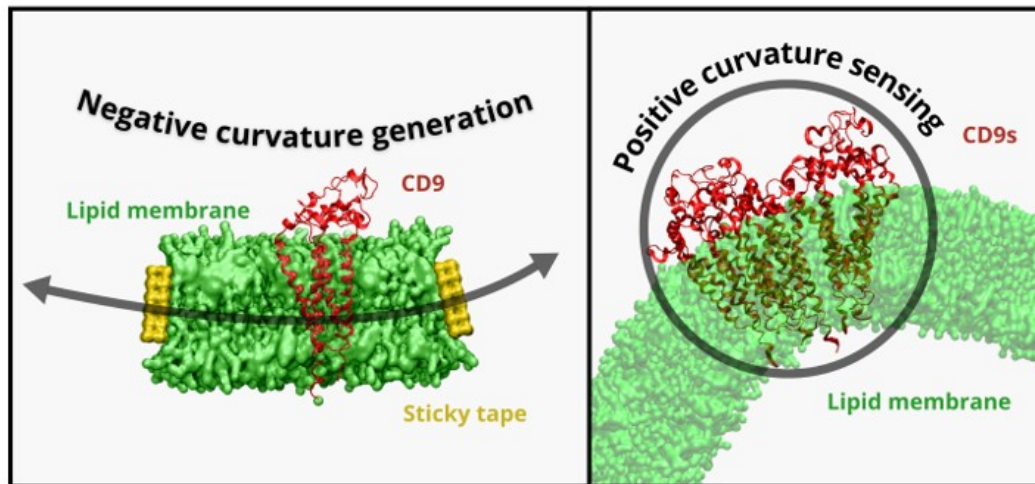


Fig. CD9 decouples curvature generation and curvature sensing.

CD9s promotes negative membrane curvature in the sticky-tape set-up, whereas preferentially partition to positively curved membrane regions. This supports a model in which CD9-induced membrane softening allows curvature generation and sensing to emerge as distinct responses.

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